Paleomagnetism of the late Paleozoic Slide Mountain terrane, northern and central British Columbia

DAVID R. RICHARDS AND ROBERT F. BUTLER
Department of Geosciences, University of Arizona, Tucson, AZ 85721, U.S.A.

AND

TEKLA A. HARMS
Department of Geology, Amherst College, Amherst, MA 01002-5000, U.S.A.

Received October 30, 1992
Revision accepted August 31, 1993

Paleomagnetic samples were collected from Mid-Pennsylvanian to Lower Permian red argillaceous cherts at two localities of the Slide Mountain terrane: 18 sites from the Sylvester allochthon in northern British Columbia and 11 sites from Sliding Mountain in central British Columbia. A secondary component of natural remanent magnetization in the Sylvester samples yields a paleomagnetic pole that can be brought into coincidence with the Jurassic portion of the North American apparent polar wander path by inferring vertical-axis rotation during obduction of the allochthon. Both localities yield a characteristic component (ChRM) with unblocking temperatures from 650 to 680°C. After structural correction for bedding tilt, all inclinations of ChRM are negative, consistent with magnetization during a reversed-polarity interval in the northern hemisphere. Site-mean ChRM directions show consistent inclinations but distinct stratigraphic groupings of declinations. Inclination-only statistics indicate that the ChRM passes a tilt test within the Sylvester allochthon and regionally between the two localities. The ChRM was apparently acquired prior to structural imbrication within the Sylvester section and regional differential tilting. We interpret the ChRM to be a primary magnetization acquired at or soon after deposition during the Permo-Carboniferous reversed-polarity superchron. The mean ChRM inclination of $-16.7° \pm 6.0°$ from the Sylvester allochthon indicates a paleolatitude of $8.8° \pm 3.4°$N, which is corroborated by a paleolatitude of $1.9° \pm 1.5°$N from the Sliding Mountain locality. When compared with expected Pennsylvanian–Permian paleolatitudes, a net poleward translation of $20.3 \pm 3.7°$ is implied for at least the sampled lithotectonic component of the Sylvester allochthon.

Des échantillons paléomagnétiques ont été recueillis dans les cherts argilacés rouges, d'âges allant du Pensylvanien moyen au Permien inférieur, à deux localités à l'intérieur du terrane de Slide Mountain, incluant 18 sites de l'allochtone Sylvester dans le nord de la Colombie-Britannique et 11 sites de Sliding Mountain au centre de la Colombie-Britannique. Une composante secondaire d'aimantation rémanente obtenue avec les échantillons du Sylvester a fourni un pôle paléomagnétique, lequel peut être amené à coïncider avec le segment jurassique sur la courbe de la dérive polaire apparente de l'Amérique du Nord, à condition que l'axe de rotation eut été vertical durant l'obduction de l'allochtone. Les deux localités fournissent une composante caractéristique (ARCa) avec températures de déblocage entre 650 et 680°C. Après correction structurale du pendage des couches, toutes les inclinaisons de l'ARCa sont négatives, ce qui est compatible avec une aimantation acquise durant un intervalle de polarité inversée dans l'hémisphère Nord. La moyenne des directions de l'ARCa des sites montre que les inclinaisons sont uniformes, mais que les déclinaisons forment des regroupements distincts. Les statistiques traitant exclusivement les données des inclinaisons révèlent que l'ARCa réussit le test de basculement pour l'allochtone de Sylvester et pour la région comprise entre les deux localités. L'acquisition de l'ARCa date apparemment d'avant l'imbrication structurale dans la section du Sylvester et du basculement différentiel. Nous interprétons l'ARCa comme étant une aimantation primaire acquise au moment du dépôt, ou peu de temps après, durant le superchrone de polarité inversée au Permo-Carbonifère. L'inclinaison moyenne de l'ARCa de $-16,7° \pm 6,0°$ pour l'allochtone de Sylvester indique une paléolatitude de $8,8° \pm 3,4°$N, laquelle est corroborée par une paléolatitude de $1,9° \pm 1,5°$N au site de Sliding Mountain. La comparaison avec les paléolatitudes appréhendées pour le Pennsylvanien–Permien, indique qu'il est nécessaire d'effectuer une translation vers le pôle de $20,3° \pm 3,7°$, qui, à tout le moins, doit être appliquée à la composante lithotectonique des échantillons de l'allochtone de Sylvester.
[Traduit par la rédaction]

Can. J. Earth Sci. **30**, 1898–1913 (1993)

Introduction

The Slide Mountain terrane of the Canadian Cordillera (Fig. 1) consists of a varied assemblage of rocks of oceanic derivation. It includes abundant radiolarian ribbon chert; featureless, probably pelagic to hemipelagic argillite; pillow and massive basalt; volcaniclastic rocks and mafic intrusive rocks; along with lesser amounts of carbonate, ultramafic rock, serpentinite, and arenite (Gordey et al. 1982; Harms 1986*a*; Monger and Ross 1971; Nelson and Bradford 1987; Struik and Orchard 1985). Where dated, units of the Slide Mountain terrane are predominantly Mississippian through Permian in age, although rocks as old as Devonian and as young as Triassic are known (Gordey et al. 1982; Harms 1986*a*; Struik and Orchard 1985).

The Slide Mountain terrane occurs as a series of discontinuous, very large scale klippen that lie along the length of the Canadian Cordilleran orogen (Wheeler et al. 1991) (Fig. 1). One of the larger of these sheet-like bodies in northern British Columbia is known as the Sylvester allochthon (Gabrielse 1963; Gordey et al. 1982), another underlies Slide Mountain in central British Columbia (Struik 1988). In its numerous localities, the Slide Mountain terrane directly structurally overlies either ancestral North American Paleozoic miogeoclinal strata, displaced miogeoclinal strata, as in the Cassiar terrane, or continentally derived Proterozoic to Paleozoic clastic strata, such as those of the Cariboo and Barkerville terranes (Wheeler et al. 1991). Its oceanic character stands in marked contrast by this juxtaposition; the Slide Mountain terrane is the most inboard of the Canadian Cordillera's noncontinental terranes.

Whereas the fundamental oceanic character of the Slide Mountain terrane is widely acknowledged, the paleogeographic

Printed in Canada / Imprimé au Canada

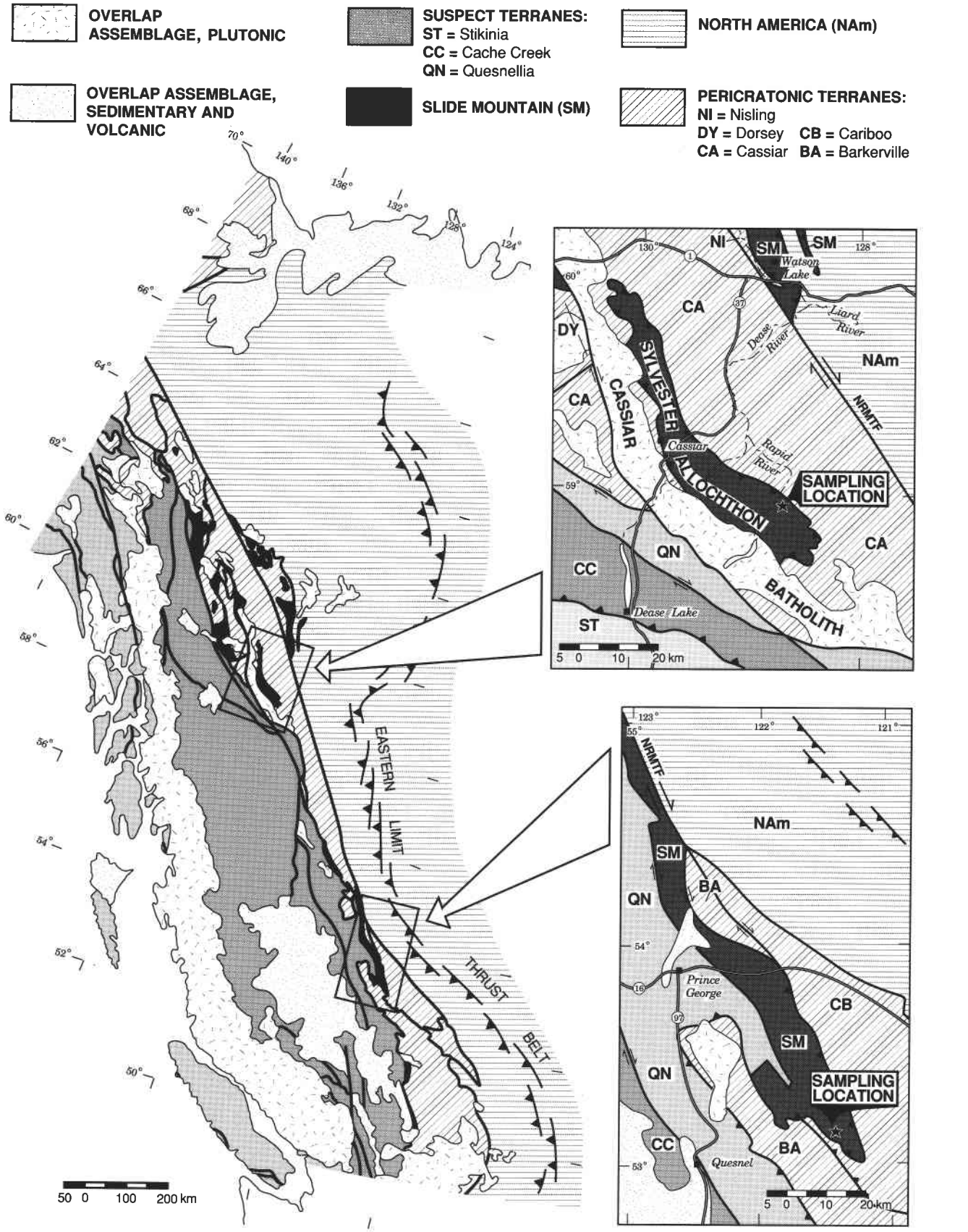


FIG. 1. Regional geology and location of the Slide Mountain terrane within the Cordillera in British Columbia and Yukon. Inset maps show the geology and location of the two paleomagnetic sampling localities in two allochthons of the Slide Mountain terrane. Adapted from Wheeler et al. (1991).

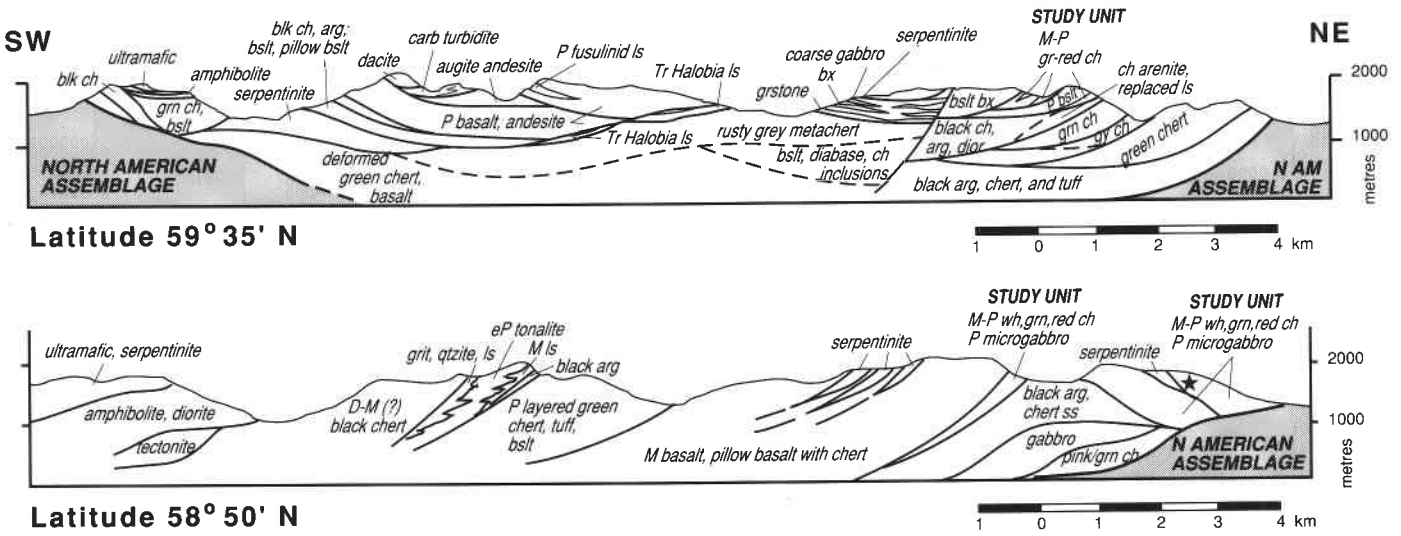


FIG. 2. Composite structure sections across the Sylvester allochthon at approximately 59°35′N (top section) and 58°50′N (lower section) showing the characteristic juxtaposition of lithotectonic slices. The position of the paleomagnetic sample locality within the Sylvester allochthon is indicated with a star. Adapted from Harms (1985*a*) and Harms et al. (1988). Horizontal and vertical scales equal.

setting for formation of its units has been the subject of some debate. Interpretations cover a spectrum from a broad proto-Pacific Ocean (Harms 1986*a*; Harms 1986*b*), based on the diverse nature of contemporaneous lithologies within the terrane, to a marginal back-arc basin (Hansen 1990; Nelson and Bradford 1989; Nelson et al. 1989), based on the presence of both volcanic arc assemblages and sedimentary rocks with North American affinity within the terrane. Many reconstructions of elements in the Canadian Cordillera during late Paleozoic time depict the Slide Mountain terrane as an ocean of indeterminate width and place minimal emphasis on the paleogeographic implications of the terrane (for example, Davis et al. 1978; Monger 1989; Struik 1987). At present, the question of whether the terrane formed largely adjacent to North America or whether it instead records the existence of a much larger ocean must be considered open (Monger et al. 1991). The resolution of this issue has broad implications for the interpretation of other Paleozoic rocks of arc, oceanic, or continental character that presently lie outboard of the Slide Mountain terrane in the Cordillera. By paleomagnetic techniques, we have interpreted the latitude of formation of one unit of the terrane that occurs both within the Sylvester allochthon and on Sliding Mountain.

Regional geology of the Slide Mountain terrane

Rather than being an orderly ophiolitic sequence, the oceanic rocks of the Slide Mountain terrane occur within innumerable fault-bounded slices, each with a characteristic internal stratigraphy or lithic suite and most with no evident genetic relationship to any other (Harms 1986*a*; Struik and Orchard 1985). Few, if any, geologic correlations can be drawn between slices. Consequently, each fault-bounded slice constitutes an independent lithotectonic component of the terrane and serves as the most appropriate unit for geologic subdivision (Harms 1986*a*). Faults that bound slices of the Slide Mountain terrane generally lie parallel to layering within each slice, but they converge along and across strike so that intervening lithotectonic units are pinched out. This results in an overall lensoidal shape for the slices, with dimensions that range from only tens of metres to one or two kilometres in thickness and a few kilometres to tens of kilometres in length along strike. With respect to one another, slices of the Slide Mountain terrane occur in an interleaved stack, where tapered ends of some nest over or under the thicker midriff of others. This structural style of the Slide Mountain terrane is demonstrated by two representative structural cross sections across the Sylvester allochthon, shown in Fig. 2.

These fault-bound slices impart a characteristic large-scale architecture to the Slide Mountain terrane that is a fundamental part of its tectonic character. This architecture illustrates that convergence and telescoping have played a major role in the construction of the terrane. Although rocks of the Milford Group in southern British Columbia span a similar age range and include some similar oceanic lithologies, they do not share this distinctive structural style (Klepacki and Wheeler 1985), and they therefore cannot share all, if any, of the structural evolution of the Slide Mountain terrane. The Milford Group is now considered part of the Kootenay terrane, a Paleozoic pericratonic terrane (Wheeler et al. 1991), rather than part of Slide Mountain terrane as it has been in the past (Monger 1977; Monger and Berg 1987). This point is critical in so far as the conclusions of this study would conflict with Paleozoic depositional links demonstrated between the Milford Group and the North American craton (Klepacki et al. 1985) if that linkage were to be extrapolated to Slide Mountain terrane.

Thrusting within the Sylvester allochthon, and by inference throughout the Slide Mountain terrane, probably concluded before or with obduction of the terrane because the basal fault of the allochthon is not broken by any significant subsequent faults. The time of obduction of the terrane is directly bracketed only by the age of the youngest unit included in the terrane (Late Triassic) and by the Cassiar batholith (mid-Cretaceous), which intrudes across the terrane boundary stitching the Sylvester allochthon to the underlying pericratonic Cassiar terrane. Regional relationships between outboard terranes and well-timed deformation within the continental margin, however, indicate that the Middle Jurassic is the most likely time of emplacement of the terrane onto the North American margin (Gabrielse 1991; Monger 1989).

Geology of sampled areas

Paleomagnetic analysis was conducted on red chert in two widely separated lithotectonic slices of the Slide Mountain terrane, one in the Sylvester allochthon and one on Slide Mountain

(Fig. 1). In both cases, the red chert occurs as part of a larger assemblage of rocks that correlate in every aspect between the two localities (Struik and Orchard 1985; Harms 1986*a*). We interpret the two sample areas, therefore, as disjunct slices of the same depositional sequence. The assemblage is dominated by a thick (108 m in a measured section on Slide Mountain (Struik and Orchard 1985) and approximately 150–200 m in the Sylvester allochthon (Harms 1986*a*)), continuous, banded radiolarian chert succession that is white to grey at its base, grades upward into olive green chert, and is capped by the interbedded brick-red or maroon chert and mudstone sampled in this study. In the Sylvester allochthon, this succession is more complete, locally including thin black chert at the base and layers of green lapilli tuff interbedded with red chert at the top. In the Sylvester allochthon and at Slide Mountain the chert sequence is intruded by microgabbro. This homogeneous, fine-grained intrusive occurs most commonly as minor dikes, sills, and pods, but locally forms small plutons in the Sylvester allochthon. Diffuse contacts between the microgabbro and red chert, in contrast to sharper contacts with underlying chert strata, and the presence of interlayered lapilli tuff at the top of the Sylvester assemblage suggest that intrusion of the microgabbro occurred with the close of chert deposition, while the red chert remained poorly lithified. Paleomagnetic sample sites were chosen in areas with little or no intrusive material.

Both chert sequences have yielded abundant microfauna. Well-preserved, diagnostic radiolarians occur in the Sylvester allochthon succession (Harms 1986*a*; Harms and Murchey 1992), while conodonts (Struik and Orchard 1985) and less well preserved, less diagnostic radiolaria (T.A. Harms, unpublished data) have been recovered from Slide Mountain. Each complete chert sequence is independently dated, and both range from Late Mississippian through Early Permian. Note that uninterrupted, quiet-water chert deposition in this one sequence spans most of the age range represented by the other units of the Slide Mountain terrane. The red chert of both sequences is Mid-Pennsylvanian through Early Permian in age, specifically, radiolarians in the Sylvester sequence require that the red chert span not less than late Des Moinesian through early Wolfcampian time, and may cover as much as Atokan through Wolfcampian time (an absolute age range of ~305-275 Ma using the time scale of Harland et al. (1990)). As the red chert section is only ~40 m thick in the Sylvester locality, and <10 m thick on Slide Mountain, it must have accumulated very slowly.

Bedding in the chert sequences, including the red chert, is readily apparent owing to the banded nature of the chert. The chert occurs in a right-way-up (demonstrated by the microfossils contained within it), generally homoclinal stack. In the Sylvester allochthon, the chert and microgabbro lithotectonic unit is repeated against itself along a large-scale, layer-parallel thrust fault. The red chert is also folded by rare, ≤1 m amplitude, overturned, chevron, intrafolial folds that suggest intrastratal slip accompanied the thrust faulting. At outcrop scale, structure is mostly limited to ≤30° variations in bedding inclination observable within the section. The time of thrust faulting in the Sylvester chert sequence is constrained only to be postdeposition (Permian) to synobduction (Middle Jurassic?).

Tectonism has resulted in a subdued thermal imprint on the chert of the sampled areas. Regionally, conodont alteration indices (CAI = 5 and, rarely, 6) in both the Sylvester and Slide Mountain allochthons (Orchard and Irwin 1988; Struik and Orchard 1985; M.J. Orchard, personal communication) indicate temperatures did not exceed ~325–450°C (Rejebian et al. 1987) throughout the history of the terrane. Although those at Sliding Mountain are less so, the quality of radiolarian preservation in the Sylvester slice is consistent with these moderate temperatures. Siliceous radiolarians should undergo significant recrystallization at any higher temperature.

Paleomagnetic sampling and analysis

Sampling localities and procedures

Paleomagnetic samples collected from the red- and maroon-banded cherts at the Sylvester and Slide Mountain localities were 2.5 cm diameter cores drilled in situ with standard paleomagnetic coring equipment. Cores were oriented by magnetic compass and inclinometer, with some orientations confirmed by sun compass.

In the Sylvester allochthon (Fig. 1), samples were collected at a locality near the Rapid River (59.0°N, 231.3°E). Preliminary samples were collected throughout the full stratigraphic thickness of Mississippian to Lower Permian white-, green-, and red-banded cherts, and from a microgabbro sill. Only the red chert was found to contain a natural remanent magnetism (NRM) of sufficient intensity and stability for paleomagnetic analysis. In subsequent detailed sampling, an average of eight cores were collected from each of 17 paleomagnetic sites in ~40 m of section at the Rapid River locality. Each paleomagnetic site was from a single chert bed of 5–12 cm thickness (at both localities). Bedding orientations vary within some of the paleomagnetic sites, as well as between sites, and local measurements were made as necessary for structural correction. These variations in attitude allowed a within-stack tilt test to help constrain the ages of components of NRM.

At Sliding Mountain (Fig. 1; 53.15°N, 238.52°E), preliminary samples were also collected from the Upper Mississippian to Lower Permian white- and green-ribbon chert, banded maroon cherty mudstone, and diabase sills. Only the maroon cherty mudstone contained NRM of sufficient intensity and stability for paleomagnetic analysis. Eight or more cores were collected from 11 paleomagnetic sites within a 3 m stratigraphic interval of the maroon cherty mudstone. The sampled section has only minor variations in bedding attitude with no internal faults.

Traces of drill barrel metal in some cases were found on the cores owing to the hardness of the cherts. Sample preparation included dipping all cores in a HCl bath to remove all traces of metal. In general, only one specimen was cut from each core. All paleomagnetic specimens were stored, measured, and demagnetized in a magnetically shielded room with field intensity <200 nT. NRM was measured with a two-axis cryogenic magnetometer (ScT C-102). Thermal demagnetization was performed in a shielded furnace with magnetic field <3 nT in the specimen region.

Rock magnetism

Typical results of coercivity-spectrum analysis are shown in Fig. 3. This analysis examined the acquisition and subsequent thermal demagnetization of isothermal remanent magnetism (IRM) (Dunlop 1972; Lowrie and Heller 1982). IRM was gradually acquired up to the maximum available magnetizing field of 700 mT. Subsequent thermal demagnetization of IRM showed unblocking temperatures distributed up to the 680°C Néel temperature of hematite. These observations indicate that hematite is the dominant ferromagnetic mineral in chert samples from both the Sylvester allochthon and the Sliding Mountain location.

Petrographic examination of Sylvester rocks (from three sites) shows abundant disseminated hematite, with some larger hema-

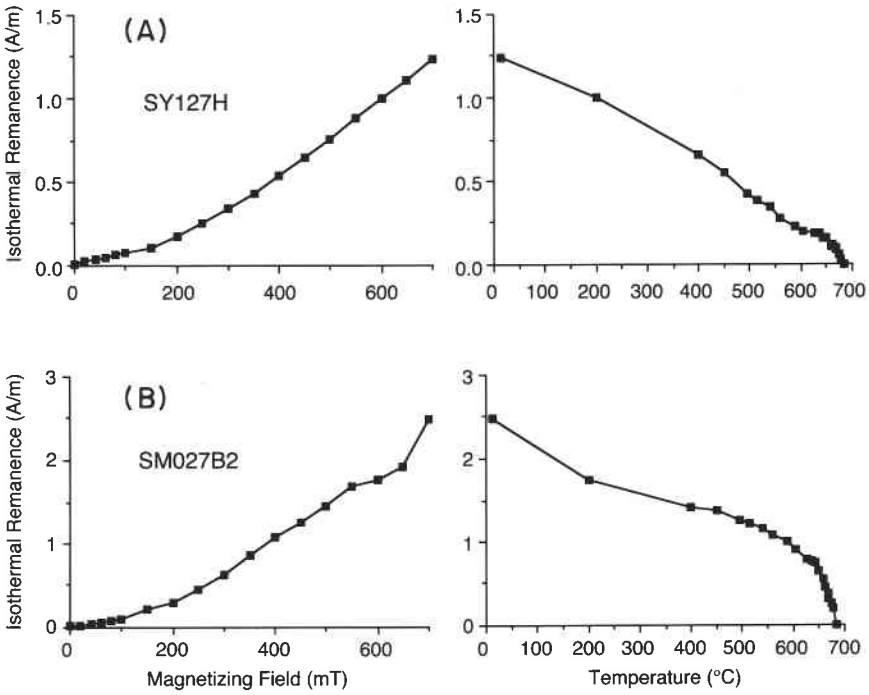


FIG. 3. Coercivity-spectrum and unblocking temperature analyses of (A) one specimen from the Sylvester allochthon and (B) one specimen from the Sliding Mountain location. In the diagrams on the left side of the figure, acquisition of isothermal remanent magnetization (IRM) is shown against magnetizing field. Thermal demagnetization of acquired IRM is shown on the right of the figure.

tite grains. These cherts are, of course, predominantly cryptocrystalline quartz in which undulose extinction was observed. Radiolaria are commonly visible (clear, without disseminated hematite) and they appear spherical. The Sliding Mountain cherty mudstone (examined from two sites) also has abundant disseminated hematite, although less than the Sylvester cherts. This cherty mudstone is predominantly cryptocrystalline quartz and clays, with undulose extinction in the quartz. Chlorite is fairly common and is typically seen along fractures, indicating that the chlorite is probably secondary. The presence of chlorite may account for the high-temperature viscous component of magnetization produced in some of the Slide Mountain samples during thermal demagnetization but never produced in the chlorite-free Sylvester samples.

Demagnetization behavior and analyses

Alternating-field demagnetization (12 steps up to 80 mT) was attempted on pilot specimens from some sites, but had no effect on the NRM of red or maroon chert specimens. Progressive thermal demagnetization at ≥ 15 temperatures from 250 to 680°C was performed on pilot specimens from all sites. The remaining specimens were thermally demagnetized at 14 temperatures from 400 to 680°C. This detailed demagnetization procedure was necessary to securely define the components of NRM. Demagnetization at temperature increments of 4–7°C between 665 and 680°C was essential to adequately determine the characteristic component of remanent magnetism (ChRM) direction for most specimens.

The vector component diagrams of Fig. 4 illustrate the range of thermal demagnetization behaviors observed for chert samples from the Sylvester allochthon. Two components of NRM are evident. A component of NRM (generally directed steeply down with declination in the southwest quadrant) displayed unblocking temperatures up to 660°C. This component will be subsequently referred to as the secondary component following arguments detailed below. The ChRM was generally revealed by thermal demagnetization between 650 and 680°C. Despite these high unblocking temperatures, detailed demagnetization experiments allowed confident determination of the ChRM direction. Approximately 90% of the specimens showed demagnetization behavior similar to that illustrated by Fig. 4A or 4B. For some specimens, such as illustrated in Fig. 4C, the ChRM composed <25% of the total NRM, and determination of ChRM direction was more difficult. Rare specimens (e.g., Fig. 4D) showed unstable behavior at high demagnetization temperatures and the ChRM direction could not be adequately determined.

The range of thermal demagnetization behaviors for specimens from Sliding Mountain locality is illustrated in Fig. 5. Again, two components of NRM are observed. A secondary, lower unblocking temperature component of NRM was removed at temperatures up to 650°C. The ChRM component was revealed by thermal demagnetization between 650 and 680°C. For most specimens, the ChRM direction could be accurately determined by the trajectory of vector endpoints (Figs. 5A and 5B). Approximately 30% of specimens acquired short-term viscous components of magnetization during thermal demagnetization to temperatures >650°C (Fig. 5C).

Principal component analysis (Kirschvink 1980) was used to determine least-squares line fits to the NRM components revealed during thermal demagnetization. For determination of specimen ChRM directions, data from a minimum of three (usually four to five) demagnetization temperatures were used for line fits in which the origin of the vector component diagram was included as an additional datum. For the secondary component, data from typically four to six demagnetization temperatures were used in principal component analysis.

For most specimens from the Sylvester allochthon, principal

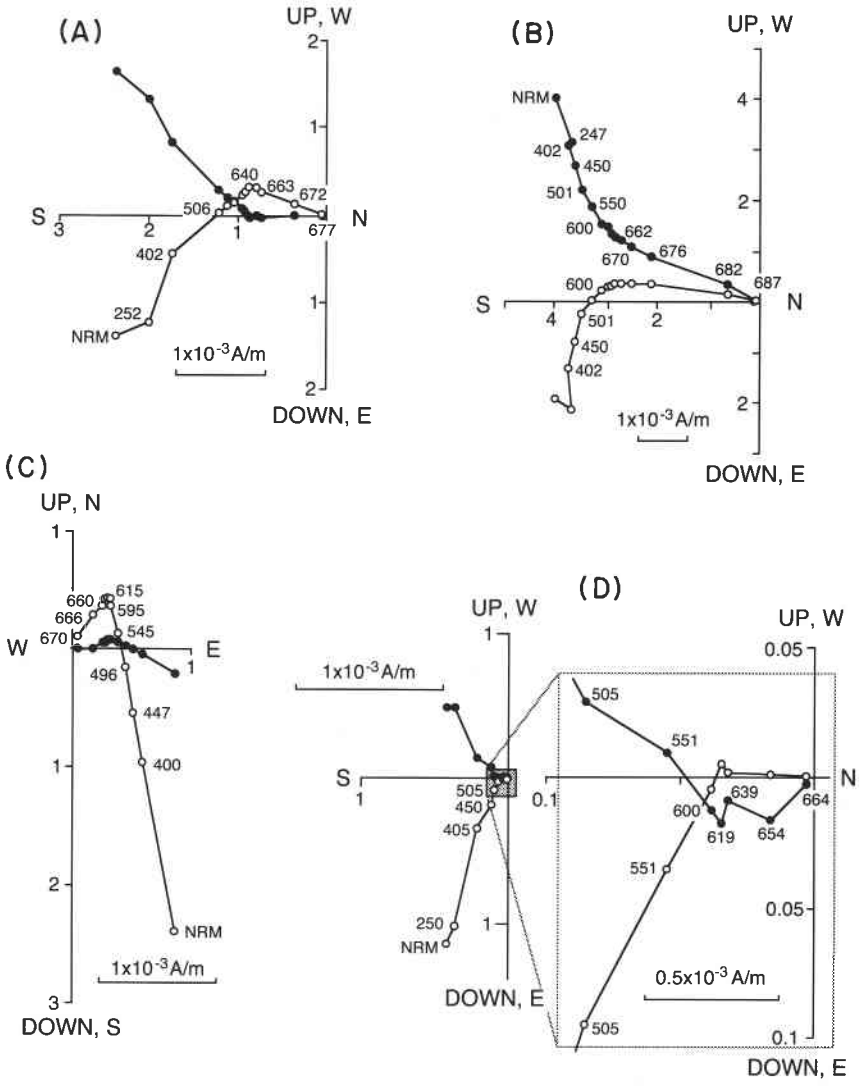


FIG. 4. Vector component diagrams of thermal demagnetization behavior of four specimens from the Sylvester allochthon. ●, projections onto the horizontal plane; ○, projections onto the vertical plane. Numbers adjacent to data points indicate temperature of thermal demagnetization in degrees Celsius. In D, an expanded diagram of the region surrounding the origin is shown to the right.

component analysis allowed determination of both the ChRM direction and the secondary direction with maximum angular deviation (MAD) $< 5°$. A few specimens displayed unstable NRM at high demagnetization temperatures (e.g., Fig. 4D) and resulting MAD $> 15°$. These specimens were rejected from further analysis. Site-mean ChRM directions and site-mean secondary directions were calculated from the respective specimen directions using standard Fisher statistics (Fisher 1953) and are listed in Tables 1 and 2. Rare samples had directions greater than two angular standard deviations from their respective site-mean directions and were excluded from the final site-mean calculations. Site-mean ChRM directions with 95% confidence limit (α_{95}) $> 15°$ are included for completeness in Table 1 but have not been included in subsequent calculations or determination of paleolatitude.

Results of principal component analysis applied to chert specimens from the Sliding Mountain locality showed that ChRM and secondary directions for most specimens were determined with MAD $< 5°$. For the few specimens that displayed viscous behavior at high temperatures (e.g., Fig. 5C), ChRM directions were retained in further analysis if a stable direction was observed over several temperature steps prior to onset of viscous behavior.

The secondary component of NRM observed for specimens from the Slide Mountain locality showed no consistency in direction either within a site or between sites. This component is not considered in further analysis. Site-mean ChRM directions and statistical parameters for sites from Sliding Mountain are listed in Table 1.

Secondary NRM component of the Sylvester allochthon

As indicated in Table 2, the secondary component extracted from the red chert of the Sylvester allochthon has well-defined site-mean directions. All 17 stratigraphic levels yield site-mean directions with $\alpha_{95} \leq 8.3°$, with average $\alpha_{95} = 5.2°$. Structural correction for bedding tilt significantly increases the scatter of the secondary directions (Figs. 6A and 6B); the estimated Fisher precision parameter (k) for the corrected directions is 18, greatly decreased from the in situ k of 63. This increase in dispersion of directions (decrease in k) is significant at the 0.1% significance level indicating that the secondary component convincingly fails the bedding tilt test. The secondary

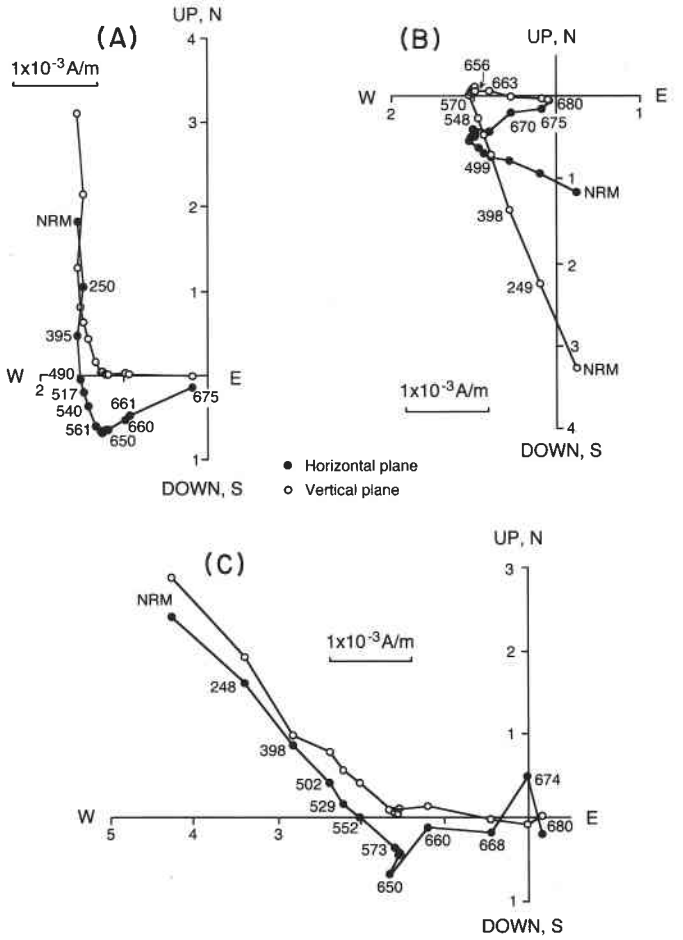


FIG. 5. Vector component diagrams of thermal demagnetization behavior of three specimens from the Sliding Mountain location. Conventions are as in Fig. 4.

component was quite clearly acquired after tilting associated with thrust stacking of the sampled section.

The mean direction of the secondary component determined from the 17 in situ site-mean directions is as follows: inclination (I_m) = 67.1°, declination (D_m) = 223.6°, with α_{95} = 4.5° (Fig. 6A). All the site-mean directions have steep downwards inclinations, indicating acquisition of the secondary component during an interval of normal geomagnetic polarity (assumes northern hemisphere). The resulting paleomagnetic pole determined from the mean of the 17 in situ site-mean virtual geomagnetic poles (VGPs) is 25.2°N, 202.4°E (95% confidence limit (A_{95}) = 6.9°, estimated angular standard deviation (S) = 15.4°).

Regional Sylvester allochthon CAI values and the thermal demagnetization behavior of the secondary component indicate that this magnetization could be of either chemical or thermal origin. CAI values of 5−6 correspond to heating to a maximum temperature of 450°C for 1 Ma or longer duration (Rejebian et al. 1987). The analysis of Pullaiah et al. (1975) indicates that heating to 450°C for ~10 Ma could produce a thermoviscous remanent magnetization with unblocking temperatures approaching 600°C. During thermal demagnetization, the secondary component is largely removed by 600°C, so a thermoviscous origin is possible, although not strongly indicated. Regional paleomagnetic and geologic evidences suggest that a chemical origin of the secondary component is also possible. We return to this topic after discussion of the possible age and implications of the posttilting pole determined from the secondary component.

Characteristic NRM component

The ChRM in both Sylvester and Slide Mountain samples have unblocking temperatures dominantly within the 650−680°C interval. CAI values and radiolarian preservation indicate that the sampled sections have not been heated to temperatures sufficient to produce a thermoviscous magnetization with such high unblocking temperatures. Therefore, the ChRM is likely a chemical remanent magnetization (CRM) carried by hematite, as has been indicated in other paleomagnetic studies of red chert sequences (e.g., Shibuya and Sasajima 1986; Hagstrum and Sedlock 1992).

The time of ChRM acquisition can be constrained to some degree by bedding-tilt tests. Site-mean ChRM directions from the red cherts of the Sylvester allochthon are shown before and after bedding-tilt correction in Figs. 7A and 7B, respectively. Structural correction improves the grouping of ChRM inclinations (shown by a locus of directions generally equidistant from the center of the plot), but the declinations remain spread over ~90°. The significance of the bedding-tilt test must therefore be evaluated using inclination-only statistical procedures such as outlined by McFadden and Reid (1982). Application of bedding-tilt corrections to the 12 site-mean ChRM directions from the Sylvester allochthon (with $\alpha_{95} < 15°$) produces an improvement in grouping of inclinations (best estimate of k increases from 9 in situ to 38 corrected, which is significant at the 0.1% significance level.

We attribute dispersion of declinations to vertical-axis rotation of layers during thrust imbrication and related flexural-slip folding of the sampled section. Indeed, groups of sites that are stratigraphically adjacent yield similar site-mean declinations (e.g., sites SY121 and SY122, see Table 1). This interpretation and the bedding-tilt test results lead to the conclusion that the ChRM was acquired prior to thrust imbrication of the sampled section. These results indicate that the chert layers within the sampled section of the Sylvester allochthon were at least coplanar, if not necessarily horizontal, at the time of ChRM acquisition (Burmester et al. 1990).

Only small variations in bedding attitude exist within the sampled section at Sliding Mountain. Site-mean ChRM directions are shown before and after bedding-tilt correction in Figs. 7C and 7D, respectively. The grouping of ChRM directions is slightly improved by application of bedding-tilt corrections, but the improvement is not statistically significant. However, a major difference in bedding attitude exists between the homoclinal tilt of the Sylvester and the Sliding Mountain sections. These differences allow a regional bedding-tilt test of the ChRM directions. If we treat all 21 site-mean ChRM inclinations from both correlatable localities as independent recordings of the paleomagnetic field in a single depositional unit, the estimated k increases from 6.6 in situ to 34 following restoration of bedding to horizontal. The increase in grouping of site-mean ChRM inclinations is significant at the 0.1% significance level. This result indicates that ChRM acquisition occurred prior to differential tilting between the two widely separated sampling localities.

After structural corrections, the mean inclination for 12 Sylvester allochthon sites is −16.7 ± 6.0°, and the mean inclination for 9 Sliding Mountain sites is −3.8° ± 2.9° (Figs. 7B, 7D; Table 3). The expected inclinations for a primary magneti-

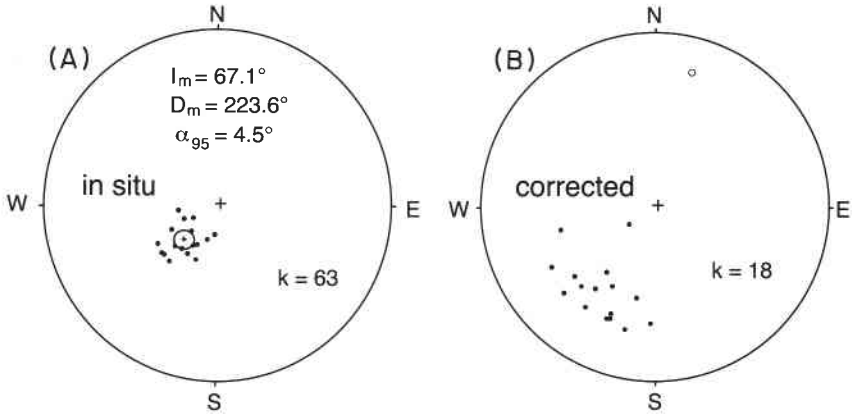


FIG. 6. Equal-area projections of site-mean directions of the secondary component from Sylvester allochthon: (A) in situ site-mean directions; (B) structurally corrected site-mean directions. Filled symbols indicate directions in the lower hemisphere; open symbols indicate directions in the upper hemisphere. Mean direction is included for in situ site-mean directions. Values of k reported are best estimates of Fisher's precision parameter.

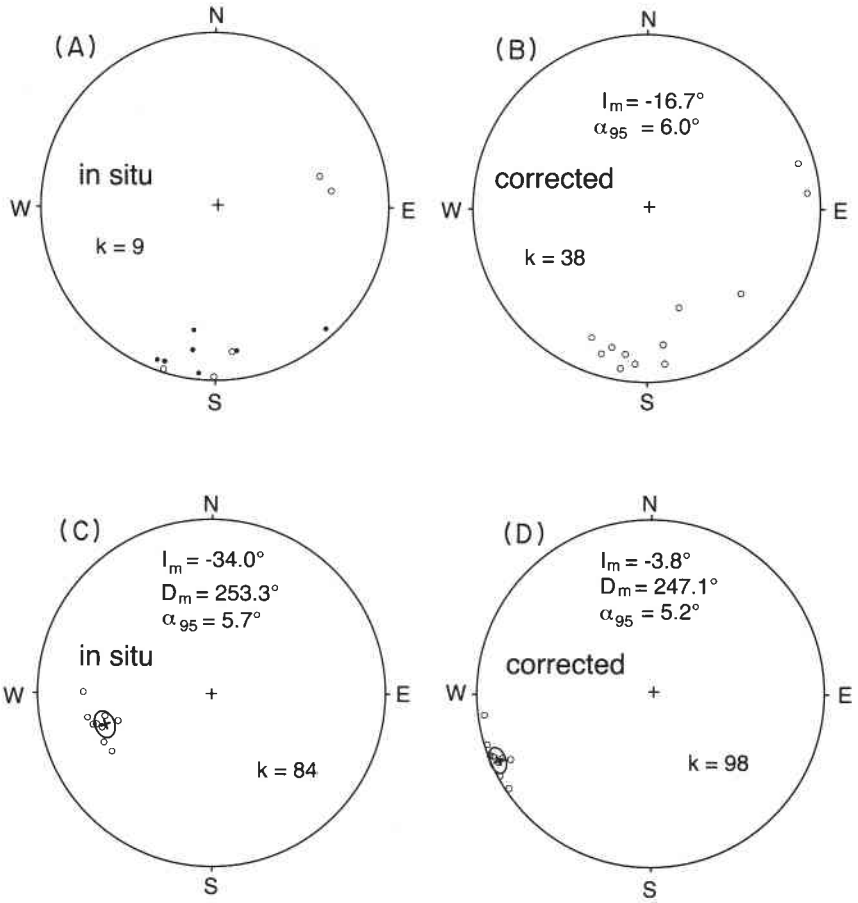


FIG. 7. Equal-area projections of site-mean directions of the characteristic component of magnetization (ChRM) from Sylvester and Sliding Mountain locations: (A) in situ site-mean ChRM directions from Sylvester allochthon; (B) structurally corrected site-mean ChRM directions from Sylvester allochthon; (C) in situ site-mean ChRM directions from Sliding Mountain; (D) structurally corrected site-mean ChRM directions from Sliding Mountain. Mean inclination is noted for structurally corrected site-mean ChRM directions from Sylvester allochthon, and mean directions are noted for Sliding Mountain. Values of k reported are best estimates of Fisher's precision parameter. Filled symbols indicate directions in the lower hemisphere; open symbols indicate directions in the upper hemisphere.

zation at the sampling locations, using a mean Late Pennsylvanian – Early Permian reference pole of 44.0°N, 124.1°E (A_{95} = 3.2°) (Van der Voo 1990), are −48.0° ± 3.7° and −39.2° ± 4.5° for Sylvester and Sliding Mountain, respectively. The observed inclinations, if primary, are clearly discordant. The mean reference pole is insensitive to age uncertainties or to the specific criteria employed because of slow apparent polar wander in the Late Pennsylvanian – Early

TABLE 1. Site-mean directions and virtual geomagnetic poles for Sylvester allochthon and Sliding Mountain characteristic component

		In situ								Corrected						
								VGP							VGP′	
Site	N/N_o	R	k	α_{95} (°)	J (10^{-3} A/m)	I (°)	D (°)	Latitude (°N)	Longitude (°E)	R'	k'	α_{95}' (°)	I' (°)	D' (°)	Latitude (°N)	Longitude (°E)
Sylvester allochthon																
SY121	8/8	7.83	41	8.8	0.4	−32.0	83.0	−11.3	336.2	7.88	60	7.2	−9.5	84.2	−1.1	328.7
SY122	7/8	6.98	314	3.4	0.8	−36.8	74.1	−9.7	345.2	6.98	266	3.7	−9.1	73.8	4.3	337.6
SY123	*9/9*	*8.07*	*9*	*18.6*	*3.6*	*5.0*	*173.5*	*−28.3*	*238.7*	*8.17*	*10*	*17.4*	*−27.2*	*174.7*	*−45.2*	*238.6*
SY124	7/7	6.95	114	5.7	1.6	3.9	185.2	−28.9	225.3	6.86	42	9.4	−10.5	184.0	−36.2	226.4
SY125	9/9	8.75	32	9.3	2.4	−15.3	173.1	−38.5	240.1	8.75	32	9.3	−39.2	161.6	−50.9	259.0
SY126	6/6	5.89	47	9.9	1.0	−0.8	180.0	−31.4	231.2	5.89	47	9.9	−21.6	172.7	−41.9	241.0
SY127	6/7	5.89	46	9.9	2.4	−1.7	197.2	−30.3	211.3	5.89	46	9.9	−18.4	193.8	−39.4	213.6
SY128	8/8	7.87	52	7.7	2.0	6.5	197.7	−26.2	211.5	7.87	53	7.7	−12.7	197.0	−35.9	210.3
SY137	*7/8*	*6.51*	*12*	*17.9*	*2.2*	*−7.5*	*101.1*	*−8.9*	*313.7*	*6.47*	*11*	*18.7*	*7.8*	*101.0*	*−2.3*	*309.8*
SY129	6/6	5.89	45	10.1	1.2	16.7	171.4	−22.1	240.5	5.89	45	10.1	−10.1	173.1	−35.8	239.8
SY135	7/7	6.85	39	9.8	2.6	5.0	137.6	−20.0	277.1	6.85	39	9.8	−27.0	132.3	−33.2	290.2
SY134	*8/8*	*6.67*	*5*	*26.7*	*3.5*	*3.6*	*187.8*	*−28.9*	*222.3*	*6.84*	*6*	*24.6*	*−13.8*	*188.6*	*−37.6*	*220.6*
SY133	8/8	7.74	27	11.0	2.5	6.2	200.5	−25.8	208.4	7.74	27	11.0	−20.0	202.8	−38.3	202.2
SY132	8/8	7.77	30	10.3	1.8	28.2	189.4	−15.6	221.9	7.77	30	10.3	−6.7	188.9	−34.0	220.5
SY131	8/8	7.87	52	7.7	2.5	17.5	188.6	−21.7	222.1	7.87	52	7.7	−16.4	188.5	−38.9	220.5
Sliding Mountain																
SM033	9/9	8.98	333	2.8	3.5	−39.6	253.9	−27.4	147.0	8.98	334	2.8	−7.6	246.2	−17.1	165.7
SM034	8/8	7.98	395	2.8	2.3	−43.5	252.6	−30.3	145.8	7.97	236	3.6	−11.6	244.3	−19.9	166.1
SM035	*7/8*	*6.36*	*9*	*20.8*	*3.7*	*−37.4*	*257.1*	*−24.3*	*145.5*	*6.37*	*10*	*20.7*	*−8.2*	*249.5*	*−15.5*	*162.7*
SM032	*7/8*	*6.37*	*10*	*20.7*	*2.5*	*−41.8*	*235.5*	*−39.6*	*161.2*	*6.37*	*10*	*20.7*	*−8.5*	*232.1*	*−25.3*	*178.0*
SM031	8/8	7.99	809	1.9	3.9	−34.4	238.9	−33.5	162.2	7.99	810	1.9	−1.5	235.9	−20.3	176.6
SM030	8/8	7.99	529	2.4	1.8	−32.6	244.6	−29.2	158.0	7.99	529	2.4	−0.7	240.9	−17.2	172.3
SM029	8/8	7.90	68	6.8	1.9	−30.6	254.3	−22.5	150.9	7.90	68	6.8	−1.1	248.6	−13.1	165.6
SM028	8/8	7.90	72	6.6	1.1	−28.4	258.4	−19.0	148.4	7.90	69	6.7	−0.7	252.0	−10.9	162.8
SM027	8/8	7.96	179	4.2	1.5	−34.7	251.9	−26.0	151.0	7.96	179	4.2	−4.3	245.5	−16.2	167.3
SM026	7/8	6.71	21	13.6	2.2	−27.7	269.9	−11.8	139.7	6.71	21	13.7	−3.8	262.3	−6.1	153.5
SM025	7/8	6.94	103	6.0	1.5	−32.4	254.1	−23.5	150.2	6.94	103	6.0	−2.7	247.9	−14.2	165.7

NOTES: Sites listed in stratigraphic order, with the stratigraphically uppermost site at the top of the table. Sites SY130 and SY136 did not have a well-defined characteristic component and were not included in determining mean direction or pole. Entries in italics have $\alpha_{95} > 15°$ and have not been used to calculate paleolatitudes. N/N_o, number of specimens used to determine site-mean direction divided by number of specimens demagnetized; R, resultant vector length of the N unit vectors; k, best estimates of Fisher's precision parameter; α_{95}, radius of cone of 95% confidence about site-mean direction; J, geometric mean intensity of characteristic component; I and D, inclination and declination of the site-mean direction; VGP, virtual geomagnetic pole. Variables marked with a prime are corrected for bedding tilt.

TABLE 2. Site-mean directions and virtual geomagnetic poles for Sylvester allochthon, secondary component

Site	N/N_o	R	k	α_{95} (°)	J (10^{-3} A/m)	I (°)	D (°)	VGP Latitude (°N)	VGP Longitude (°E)
SY121	7/8	6.95	119.7	5.5	0.5	77.2	242.7	43.1	201.1
SY122	8/8	7.98	447.4	2.6	1.2	71.5	262.6	42.5	182.8
SY123	9/9	8.87	64.0	6.5	2.5	63.5	219.5	19.0	202.9
SY124	7/7	7.00	1496.5	1.6	0.9	63.5	212.0	17.4	208.2
SY125	9/9	8.97	236.3	3.4	2.2	73.0	225.4	32.9	205.1
SY126	6/6	5.94	78.1	7.6	0.5	68.3	211.6	23.5	210.4
SY127	6/7	5.96	124.1	6.0	1.3	56.7	237.3	17.3	186.7
SY128	8/8	7.97	246.2	3.5	1.0	62.3	226.3	19.5	197.6
SY129	6/6	5.97	174.0	5.1	0.7	69.4	206.9	24.1	213.9
SY130	7/7	6.99	474.6	2.8	0.6	73.5	248.3	39.8	193.2
SY131	8/8	7.85	45.6	8.3	1.5	55.4	229.6	13.5	192.0
SY132	8/8	7.93	99.4	5.6	1.1	62.8	201.6	14.8	215.5
SY133	8/8	7.96	189.4	4.0	1.3	56.1	227.3	13.3	194.0
SY134	7/8	6.91	63.9	7.6	1.7	55.2	221.0	10.6	198.5
SY135	7/7	6.94	103.0	6.0	1.3	73.4	196.2	28.9	221.9
SY136	8/8	7.96	166.2	4.3	0.3	65.8	242.4	28.6	188.9
SY137	8/8	7.88	57.9	7.3	1.0	76.2	185.1	32.9	228.6

NOTES: See Table 1 for explanation. All values here are in situ.

TABLE 3. Mean inclination, paleolatitude, and poleward transport

	Inclination (°)[a]	Paleolatitude (°N)[b]	Expected paleolatitude (°N)[c]	Poleward transport (°)[d]
Sylvester allochthon	−16.7±6.0	8.8±3.4	29.1±3.2	20.3±3.7
Sliding Mountain	−3.8±2.9	1.9±1.5	22.2±3.2	20.3±2.8

NOTES: Error estimates listed are 95% confidence limits.

[a]Mean inclination of structurally corrected characteristic remanent magnetization (ChRM) calculated from site-mean ChRM directions using inclination only statistics (McFadden and Reid 1982).

[b]Mean paleolatitude determined by applying inclination only statistics to site-mean paleocolatitudes as described by Plumley et al. (1983).

[c]Paleolatitude of collecting location predicted from reference paleomagnetic pole at 44.0°N, 124.1°E (α_{95} = 3.2°).

[d]Latitudinal motion implied by difference between observed and expected paleolatitudes (Beck et al. 1986).

Permian and an abundance of paleomagnetic poles for this time period. Hence, the observed inclination discordance is a robust result, not subject to reference pole uncertainties. We interpret the age and tectonic significance of the ChRM from these basic results in the sections below.

Interpretations of paleomagnetic results

Secondary NRM component of the Sylvester allochthon

Calculation of the paleomagnetic pole (Fig. 8) from the secondary component of the Sylvester allochthon assumes that present horizontal approximates paleohorizontal at the time the secondary magnetization was acquired. In fact, we have only indirect information on paleohorizontal over part of the geological time interval during which the secondary NRM could have been acquired. The basal Sylvester thrust is regionally subhorizontal and is intruded by the 105 Ma Cassiar batholith. Thus no significant tilting of the Sylvester allochthon is likely to have occurred since the allochthon came to rest on the Cassiar terrane, which demonstrably precedes 105 Ma and is interpreted to be Middle Jurassic by regional geologic constraints.

Rotation and polewards transport of the Sylvester allochthon since the middle Cretaceous are also constrained by paleomagnetic and geologic evidence. Butler et al. (1988) determined a paleomagnetic pole from late Paleozoic rocks of the Sylvester allochthon that were remagnetized in the contact metamorphic aureole of the Cassiar batholith. When compared to the mid-Cretaceous reference pole for North America (Globerman and Irving 1988), the pole from the remagnetized Paleozoic rocks indicates no statistically significant vertical-axis rotation ($R \pm \Delta R = -7.4° \pm 20.5°$) and minimal polewards translation ($p \pm \Delta p = 7.9° \pm 7.8°$). Geological estimates of ~700 km of poleward translation of the Cassiar terrane since mid-Cretaceous (Gabrielse 1985) are in accordance with these results. These evidences preclude large amounts of vertical-axis rotation, polewards transport, or significant tilting of the Sylvester allochthon since mid-Cretaceous and likely since Middle Jurassic.

The paleomagnetic pole determined from the secondary component of the Sylvester allochthon (Sylvester secondary pole; Fig. 8) is far removed from the Permian through Cenozoic portion of the North American apparent polar wander (APW) path. Hence, a major structural disturbance must have occurred

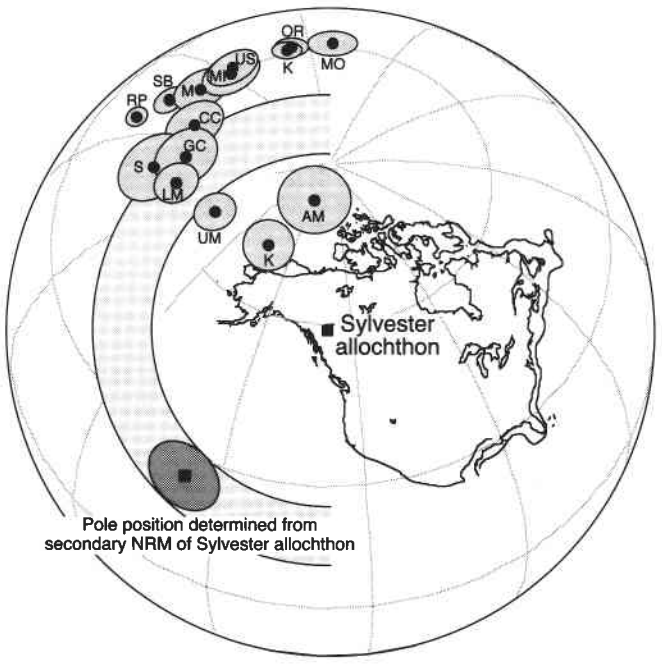


FIG. 8. Paleomagnetic pole determined from secondary magnetization from Sylvester allochthon. The orthographic projection is centered on the Sylvester allochthon. Paleomagnetic pole calculated from the in situ secondary component of magnetization from the Sylvester allochthon is shown by the square with attendant 95% confidence limits. The stippled semicircle centered on the Sylvester allochthon indicates the pole positions that could result from applying vertical-axis rotations to the allochthon. Late Triassic through Cretaceous North American reference paleomagnetic poles from the continental interior are also shown by circles with surrounding 95% confidence limits indicated by stippled ovals. RP, Red Peak Formation (~245 Ma) (Shive et al. 1984); SB, State Bridge Formation (~245 Ma) (Christensen 1974 from Gordon 1984); M, Moenkopi Formation (~240 Ma) (Helsley and Steiner 1974); MI, Manicouagan impact structure (~215 Ma) (Robertson 1967; Larochelle and Currie 1967); US, Upper Shale Member of Chinle Formation (~215 Ma) (Bazard and Butler 1991); MO, Moenave Formation (~200 Ma) (Ekstrand and Butler 1989); CC, Corral Canyon (172 Ma) (May et al. 1986); GC, Glance Conglomerate (151 Ma) (Kluth et al. 1982); LM, Lower Morrison Formation (~150 Ma) and UM, Upper Morrison Formation (~145 Ma) (Steiner and Helsley 1975); K, mid-Cretaceous average (Globerman and Irving 1988); AM, Adel Mountains pole (~75 Ma) (Gunderson and Sheriff 1991); OR, Owl Rock Member of Chinle Formation (Bazard and Butler 1991); S, Summerville Formation (~158 Ma) (Bazard and Butler 1992).

after the acquisition of this secondary component of magnetization. From the arguments presented above, the discordance between the Sylvester secondary pole and the North American APW path must have been produced before the allochthon finally came to rest on North American Paleozoic miogeoclinal strata of the Cassiar terrane. The secondary magnetization itself, therefore, must be Middle Jurassic or older.

We lack direct evidence for paleohorizontal between Late Permian and Middle Jurassic, and tectonic interpretations drawn from the secondary component are correspondingly uncertain. However, if we assume that present horizontal approximates paleohorizontal during acquisition of the secondary component (based on the extensive bedding-parallel basal Sylvester thrust) and that subsequent deformation was limited to vertical-axis rotation (and paleomagnetically undetermined longitudinal translation), an intriguing possibility becomes evident. The locus of positions of the Sylvester secondary pole resulting from vertical-axis rotation of the sampled section is a circle of latitude centered on the collecting location (Fig. 8). This locus of pole positions intersects the Jurassic portion of the North American APW path. An attractive interpretation of the secondary component of magnetization is that it was acquired in the Jurassic during the obduction of the Sylvester allochthon. Secondary magnetizations are commonly associated with regional thrusting, as has been observed throughout the Alleghenian orogen (McCabe and Elmore 1989). Subsequent vertical-axis rotation of thrust sheets resulting from continuing emplacement of the terrane and shortening in the underlying continental margin could have displaced the secondary paleomagnetic pole from the North American APW path.

This interpretation is made more attractive by the observations of Cole et al. (1992) that Paleozoic rocks of the northern Cache Creek terrane, to the west-northwest of the Sylvester allochthon, were apparently chemically remagnetized during Early Jurassic time. A regional chemical remagnetization event may have affected the northern Canadian Cordillera during the Late Triassic to Middle Jurassic. This chemical remagnetization may be associated with compressional deformation that juxtaposed the Cache Creek and Stikinia terranes and obducted the Sylvester allochthon onto the North American continental margin (Monger 1989). Remagnetization could have resulted from chemical reactions involving orogenic fluids flushed through the rocks in a manner analogous to late Paleozoic remagnetization during the Alleghenian orogeny (McCabe and Elmore 1989; Miller and Kent 1988; Oliver 1986). A possible drawback of this interpretation is that the red cherts of the Sylvester allochthon probably have very low fluid permeability, which would inhibit penetration by orogenic fluids (although the low permeability may be exactly why the older ChRM is well-preserved in these rocks). While the exact origin of the secondary component of NRM is uncertain and attendant tectonic interpretations remain speculative, acquisition of this magnetization during Jurassic obduction of the Sylvester allochthon is consistent with local and regional geologic and paleomagnetic evidence.

Characteristic NRM component

All available information suggests that the ChRM is of primary origin, acquired at or soon after deposition while the strata were horizontal. The very high temperature stability of the ChRM indicates a chemical origin, most likely the result of oxidation penecontemporaneous with deposition. The ChRM directions have consistent shallow negative inclinations within the directional dispersion expected for paleosecular variation. This observation is consistent with acquisition of the ChRM throughout Mid-Pennsylvanian to Early Permian deposition of the red chert in the northern hemisphere during the Permo-Carboniferous reversed-polarity superchron. Passage of the inclination-only bedding-tilt test within the Sylvester sequence, and also regionally between the Sylvester and Sliding Mountain localities indicates that the ChRM was acquired before the tilting both within the Sylvester section and between the two localities.

Direct control on the time of bed tilting in the sampled sections is limited. Stacking of slices within the Sylvester allochthon began before its obduction and probably continued during the emplacement process. In some areas of the allochthon, structural stacking commenced in the Permian (Harms 1985). Although

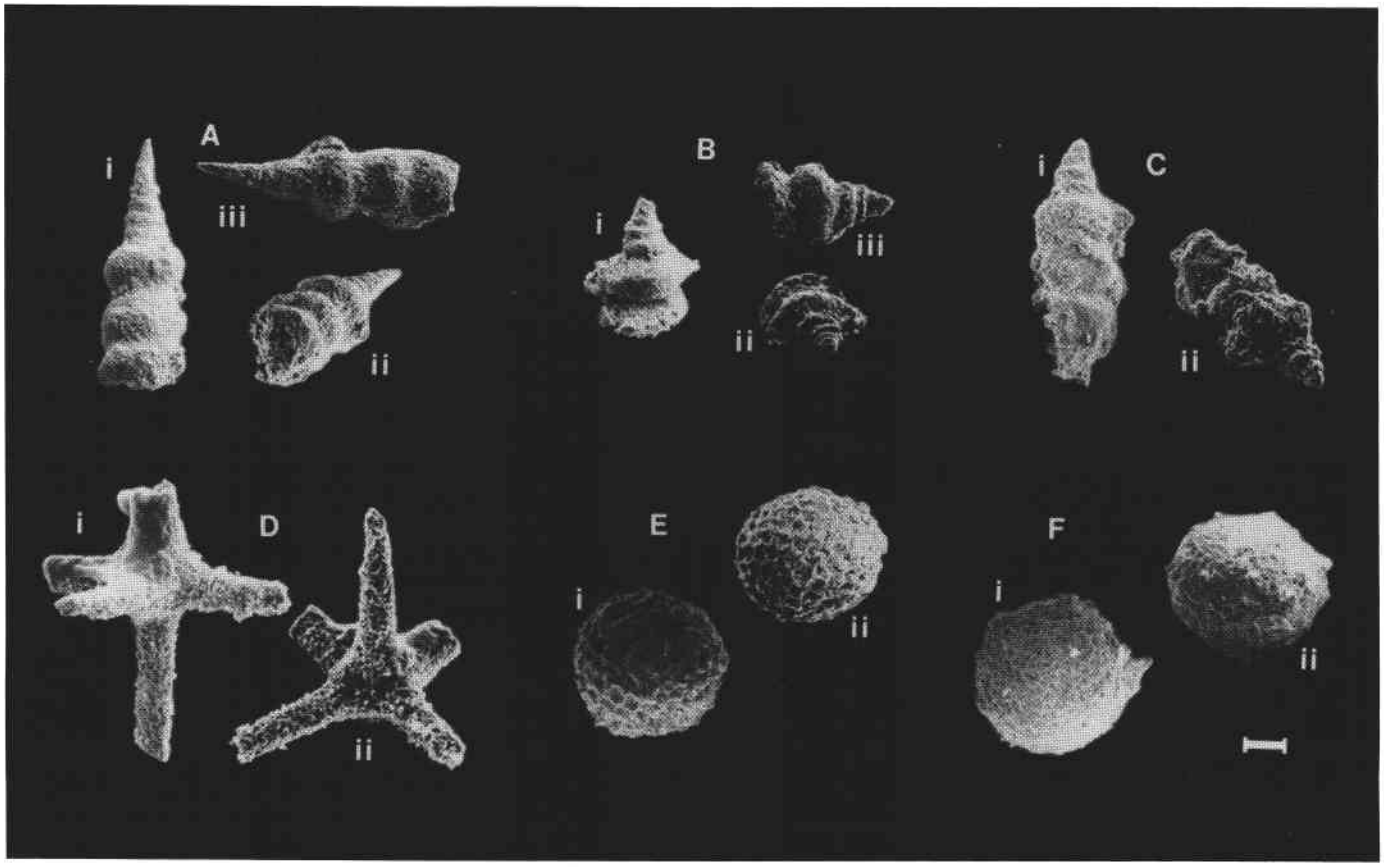


FIG. 9. Radiolarians and sponge spicules from the sampled sections in the Sylvester allochthon and on Sliding Mountain. To demonstrate the absence of deformation of the microfossils, each specimen is shown in two or three views: (*i*) vertical view of specimen; (*ii*) side view at approximately 50° from vertical; (*iii*) second side view, also at approximately 50° from vertical, and 90° (in the horizontal plane) from view *ii*. (A) *Parafollicucullus lomentaria* Ishida & Imoto: Sylvester section red chert, composite sample GAH-84-322; Wolfcampian; scale bar = 50 mm (200×). (B) *Pseudoalbaillella sp.*: Sylvester section red chert, composite sample GAH-84-322; scale bar = 50 mm (200×). (C) *Pseudoalbaillella nodosa* Ishiga: Sylvester section red chert, composite sample GAH-84-322; Atokan-Des Moinesian; scale bar = 50 mm (200×). (D) Sponge spicule: Sylvester section red chert, composite sample GAH-84-322; scale bar = 50 mm (200×). (E) Spumellarian: Sliding Mountain section red chert, sample 87-SM-85A; scale bar = 68 mm (150×). (F) Spumellarian: Sliding Mountain section red chert, sample 87-SM-85A; scale bar = 68 mm (150×).

there is no other direct control on the age of thrust imbrication of the sampled section, it could be similarly early. The most likely time of tilting, and the thrust faulting it may be related to, is not long after the close of chert deposition.

The fact that the inclinations of the ChRM directions from the two sampling locations resoundingly pass the regional bedding-tilt test requires that the two stratigraphic sequences were coplanar (horizontal or tilted by the same amount in the same direction) at the time of ChRM acquisition. It is common in paleomagnetic studies to further conclude that ChRM directions that pass such bedding-tilt tests were acquired while the strata were horizontal. However, the bedding-tilt test only directly requires that the layers were coplanar and not necessarily horizontal at the time of ChRM acquisition (Burmester et al. 1990). This may seem an exacting distinction, but it is at the heart of the paleogeographic and tectonic interpretations of the ChRM discussed below.

These observations and arguments suggest that the simplest interpretation of the ChRM is as a primary magnetization acquired while the strata were in their original horizontal configuration. Unlikely coincidences of bedding attitudes and paleomagnetic field directions are required to account for these observations by hypothesizing acquisition of ChRM after some portion of the presently observed tilting had occurred. Barring effects (e.g., compaction) that could introduce systematic errors in the ChRM directions, we believe the ChRM directions can be used to infer the original paleolatitudes of sampled Sylvester and Sliding Mountain red chert sequences.

Tectonic discordance or compaction causing shallower than expected inclinations?

Our study yields characteristic paleomagnetic inclinations from both study areas that are shallower than the expected inclinations by similar amounts. Could this systematic discordance have arisen from an inherent, uniform distortion of the bedded cherts sampled, such as compaction flattening during diagenesis, rather than through latitudinal translation of this unit? We do not find that enough is known about diagenesis of banded chert sequences to adequately address questions regarding chert compaction (see, for example, Hesse 1988; Jones and Murchey 1986; Murray et al. 1992 and references therein). However, the radiolarian microfossils within the cherts of this study are reliable strain markers (Toriumi 1982, 1986; Toriumi et al. 1988) and thus provide direct evidence concerning any change in shape of the chert sequences.

Parafollicucullus lomentaria Ishiga & Imoto (Fig. 9A),

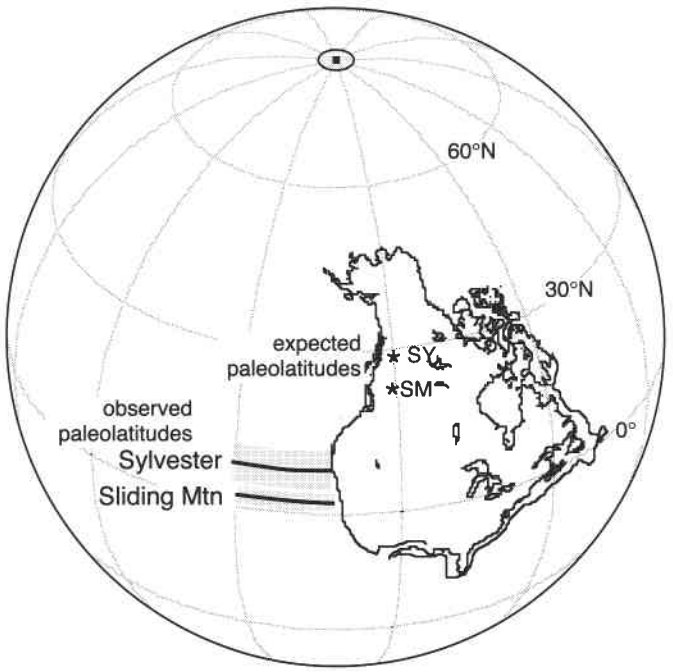


FIG. 10. Paleogeographic positions of the Sylvester allochthon and Sliding Mountain in the Late Pennsylvanian – Early Permian. The paleomagnetic reference pole for the Late Pennsylvanian – Early Permian is used as the pole of the paleogeographic grid. The observed paleolatitudes determined from the paleomagnetic directions for the Sylvester allochthon and for Slide Mountain are shown by the heavy lines with confidence limits shown by the stipple bands of latitudes. Present positions of the Sylvester allochthon and Slide Mountain within the Canadian Cordillera are shown by SY and SM, respectively.

Pseudoalbaillella sp. (Fig. 9B), and *Pseudoalbaillella nodosa* Ishiga (Fig. 9C), found in the Sylvester red chert, are constructed of combinations of spherical shapes and can thus function as strain spheroids, as can simple, corroded, spineless spumellarians (nondiagnostic spheres) that are abundant in the Sliding Mountain chert (Figs. 9E, 9F). Complex shapes of sponge spicules with rods in three perpendicular directions (Fig. 9D), found in Sylvester red chert, are also ideal strain indicators. Because there is no compositional difference (only a crystallinity difference) between them, the radiolarians and sponge spicules should deform uniformly, or nearly so, with the siliceous matrix in which they occur, as has been successfully tested by Toriumi et al. (1988). So, in form and in rheology, microfossils provide the necessary attributes to assess any changes in shape the chert as a whole may have suffered.

Figure 9 shows clearly that microfossils within the cherts of the sampled sections are undistorted. We suggest that this observation can be applied to the chert beds as a whole, and that it provides good evidence against compaction and paleomagnetic inclination flattening. We believe the paleomagnetic inclination discordance, therefore, is the result of post-Early Permian poleward translation.

Paleogeographic and tectonic implications

Determination of paleolatitude and polewards transport

To determine the paleolatitude of origin of the two sampled stratigraphic sections of the Slide Mountain terrane, we followed a modification of the procedure suggested by Plumley et al. (1983). No correction to the ChRM directions for the displacement recorded by the secondary component is necessary, as our interpretation of the ChRM component relies solely on its inclination. Site-mean ChRM directions are taken as independent recordings of the paleomagnetic field penecontemporaneous with deposition of each chert bed. The paleocolatitude, p_0, for each site yielding a site-mean ChRM direction was determined as the angular distance from the sampling location to the corresponding site-mean virtual geomagnetic pole (listed in Table 1). The apparent northern hemisphere paleolatitude of each site was then determined as $p_0 - 90°$ because all ChRM directions are interpreted to have been acquired during the Permo-Carboniferous reversed-polarity superchron. The dispersion of apparent northern hemisphere paleolatitudes between sites is attributed to sampling of paleosecular variation. Average paleolatitudes for each of the two chert sequences were determined separately by applying the statistical procedure of McFadden and Reid (1982) to the apparent paleolatitudes. The Sylvester allochthon section yields a paleolatitude of deposition of $8.8 \pm 3.4°$N, and the Sliding Mountain section one of $1.9 \pm 1.5°$N. Full results of these calculations are summarized in Table 3.

Based on a Late Pennsylvanian to Early Permian age of magnetization, observed paleolatitudes of $8.8° \pm 3.4°$N and $1.9° \pm 1.5°$N for the Sylvester section and Sliding Mountain section, respectively, show latitudinal discordance of $\sim 20°$ to the south with respect to each of their present positions (Richards et al. 1991) (Fig. 10). The age of ChRM magnetization, within the Permo-Carboniferous reversed-polarity superchron, removes hemispheric ambiguity from these results. The North American reference pole used to determine expected Permo-Carboniferous paleolatitudes for the two Slide Mountain terrane study sites is an average of 14 Upper Carboniferous through Lower Permian poles listed by Van Der Voo (1990), with a quality factor (Q) ≥ 5. As discussed above, because of slow apparent polar wander and an abundance of paleomagnetic poles available for calculation of a mean reference pole, the resulting reference pole is quite insensitive to age uncertainties or to the specific selection criteria employed. This reference pole and expected and observed paleolatitudes for both study sites are listed in Table 3.

At face value, the paleolatitudes calculated from the ChRM directions of the Sliding Mountain and Sylvester allochthon sections are statistically distinct. However, we caution against such a literal interpretation of these results. The stratigraphic section of red cherts in the Sylvester allochthon is of significant thickness and almost certainly has afforded adequate temporal averaging of geomagnetic paleosecular variation. However, the sampled section at Sliding Mountain from which ChRM directions were isolated is only 3 m thick, and there is much less certainty that paleosecular variation has been adequately sampled. The paleomagnetic data from Sliding Mountain are not sufficiently robust to stand alone. The value of the paleomagnetic results from Sliding Mountain is that they corroborate, through the regional bedding-tilt test, the results from the Sylvester allochthon, which provide a much more accurate estimation of paleolatitude.

Tectonic implications

In the strictest application, because of the internal architecture of genetically-unrelated fault-bounded lithotectonic slices that characterizes the Slide Mountain terrane, our paleomagnetic results only constrain the paleogeography of the chert and microgabbro sequence analyzed. They do not necessarily apply

to the Slide Mountain terrane as a whole. Indeed, some lithotectonic units of the terrane resemble distinctive and coeval strata of the distal, northern Cordilleran miogeocline (Harms 1986*a*; Nelson et al. 1988) and probably have not undergone significant latitudinal or longitudinal transport with respect to cratonic North America. At a minimum these results do, however, demonstrate that material has been incorporated into the Slide Mountain terrane from a great distance. This substantiates the hypothesis that the structure of the terrane is the result of significant telescoping of Paleozoic oceanic crust (Harms 1986*b*), at least between those slices that are far traveled and those that can be correlated to the northern North American continental margin.

Our results are in accord with models that consider the Slide Mountain terrane as a relic of the ocean basin that extended to the west from the distal Paleozoic passive margin of North America, as part of the North American plate (e.g., Monger et al. 1991, and references therein). Our results are in conflict with models that represent the paleogeography of the Slide Mountain terrane as a marginal basin. For example, Nelson et al. (1988) place the chert sequence of the Sylvester allochthon study area in their "Division II," a subdivision of the Sylvester allochthon that they interpret as a back-arc basin.

Our interpretation of the paleomagnetism of the red chert sequence indicates the following tectonic history for the Mississippian through Permian chert and microgabbro lithotectonic unit of the Slide Mountain terrane:

(1) Red chert deposition commenced in the Mid-Pennsylvanian in tropical waters, ~20° south of the present position of these rocks with respect to North America. Low paleolatitudes of formation are consistent with an occurrence of the radiolarian *Follicucullus monacanthus* Ishiga & Imoto in the same red chert unit in another, separate slice in the northern Sylvester allochthon (Harms and Murchey 1992), a form which appears to have been restricted to tropical latitudes (Murchey and Jones 1983). Deposition of the red chert ended in the Early Permian with an episode of minor explosive volcanism and high-level intrusion of microgabbro.

(2) Northward drift of the unit occurred during Late Permian to Middle Jurassic. This requires a minimum average rate of approximately 1.5 cm/year based on 14° latitudinal displacement (~20° paleomagnetic discordance minus ~6° maximum displacement on Late Cretaceous – Early Tertiary dextral transcurrent faults). Any longitudinal component of motion is, of course, undetermined, as is the case with all paleomagnetic studies. Northward drift must have occurred with, or as a result of, telescoping of upper ocean crust and disposal of lower oceanic crust and lithosphere. The unit was incorporated into the Slide Mountain terrane sometime after the Permian and before the Middle Jurassic. Internal imbrication and homoclinal tilting of the unit probably occurred at this time. The allochthons of the terrane were subsequently accreted onto the North American continental margin.

(3) The secondary component of magnetization suggests that flushing of orogenic fluids from the zone of convergence accompanied Middle Jurassic obduction of the Slide Mountain terrane, causing a partial chemical remagnetization of the chert sequence at that time.

(4) Continued convergence between the Slide Mountain terrane and the craton caused the rotation of the allochthons about a vertical axis, producing discordance in the declination of the secondary component.

The discordance of inclination in the ChRM component of the red chert sequence, and the late Paleozoic history which that magnetization indicates, are constraints that must be considered in paleogeographic reconstructions of other lithotectonic units that constitute the Slide Mountain terrane *and* in late Paleozoic reconstructions of terranes that now lie outboard of the Slide Mountain terrane. The latter includes, in particular, the Yukon–Tanana, Nisling, and Dorsey terranes, the basement of Quesnellia and Stikinia, and parts of the Cache Creek terrane. Only one other paleomagnetic study bears on the late Paleozoic paleogeography of the terranes in the Intermontane region: the Early Permian Asitka volcanic rocks from Stikinia yielded concordant paleolatitudes from inclination data that are pretilting and apparently primary (Irving and Monger 1987). As noted by the authors, the Asitka results, while of excellent quality, are preliminary because of the small number of samples and sites. While interpretations of the late Paleozoic paleogeography of the Intermontane terranes abound, we are only beginning to acquire late Paleozoic paleomagnetic results and to understand their implications for interpretations of the paleogeography. However, one important consistency between the Asitka results and observations from the Slide Mountain terrane is worthy of special emphasis. Both results clearly indicate that the Intermontane terranes originated in the northern hemisphere. Further discussions of Mesozoic paleomagnetic results from the Intermontane terranes are logically confined to northern hemisphere paleolatitudes as noted by Vandall and Palmer (1990).

Summary and conclusions

We have determined a well-defined characteristic component of NRM from correlative Sylvester allochthon and Sliding Mountain red chert units, and a well-defined secondary component of NRM from the Sylvester unit. These paleomagnetic data require two distinct structural "disturbances." The ChRM is older than tilting associated with thrust stacking of the banded chert unit at the Sylvester locality and older than differential tilting between the two allochthons. The secondary component at the Sylvester locality must be younger than the small-scale internal folding and thrust imbrication of the sampled section, but the chert unit as a whole must have been disturbed later because the secondary pole falls far from the North American APW path. Regional geologic and previous paleomagnetic data (Butler et al. 1988; Gabrielse 1985) establish that "disturbances" after the allochthon came to rest in the Middle Jurassic (maximum 6° northward translation, negligible rotation, negligible tilting) cannot account for the displacement of the secondary component pole from the North American APW path. Therefore, the secondary component was acquired before the Sylvester allochthon came to rest on the North American miogeoclinal strata of the Cassiar terrane in the Jurassic. The simplest interpretation is that this secondary component resulted from chemical reactions involving orogenic fluids during Early Jurassic obduction of the Sylvester allochthon. Following acquisition of this secondary component of magnetization, the sampled lithotectonic unit was rotated about a vertical axis before it finally came to rest on the Cassiar terrane.

The probable Jurassic acquisition of the secondary component combined with the pretilting age of the characteristic component of magnetization requires that the ChRM by pre-Jurassic and older than the thrusting events known from this unit of the Slide Mountain terrane. The pretectonic timing required for the ChRM, the single polarity, consistent with a

primary magnetization of these Late Pennsylvanian – Early Permian rocks, and the high temperature stability, strongly argue that the ChRM is a primary magnetization that conveys original paleolatitude information. In this preferred interpretation, the red chert unit of the Slide Mountain terrane has been translated poleward by ~20° relative to North America, with most of this translation between Early Permian and Jurassic. Since the regional tilt test only requires that the sampled sections were coplanar when the ChRM was acquired, a more complex interpretation involving secondary acquisition of ChRM and without resulting translation information is technically possible but not preferred.

Acknowledgments

Critical logistical field support during the summers of 1987 and 1988 was provided by the Geological Survey of Canada through H. Gabrielse and the Canada – British Columbia Mineral Development Agreement through J.-A. Nelson of the British Columbia Ministry of Energy, Mines and Petroleum Resources. Their help is greatly appreciated. L.C. Struik kindly introduced us to the geology of Sliding Mountain. We thank N. Meader for help with tables and final checking of the paper. This research was supported by PRF grant #18432-AC2 and a Summer Research Fellowship ride-on grant to T.A. Harms.

Bazard, D.R., and Butler, R.F. 1991. Paleomagnetism of the Chinle and Kayenta formations, New Mexico and Arizona. Journal of Geophysical Research, B, **96**: 9847–9871.

Bazard, D.R., and Butler, R.F. 1992. Paleomagnetism of the Middle Jurassic Summerville Formation, East Central Utah. Journal of Geophysical Research, B, **97**: 4377–4385.

Beck, M.E., Jr., Burmester, R.F., Craig, D.E., Grommé, C.S., and Wells, R.E. 1986. Paleomagnetism of middle Tertiary volcanic rocks from the Western Cascade series, northern California: Timing and scale of rotation in the southern Cascades and Klamath Mountains. Journal of Geophysical Research, B, **91**: 8219–8230.

Burmester, R.F., Bazard, D.R., and Beck, M.E., Jr. 1990. Post-folding remagnetization that passes the fold test. Geophysical Journal International, **102**: 455–463.

Butler, R.F., Harms, T.A., and Gabrielse, H. 1988. Cretaceous remagnetization in the Sylvester Allochthon: limits to post-105 Ma northward displacement of north-central British Columbia. Canadian Journal of Earth Sciences, **25**: 1316–1322.

Christensen, F.D. 1974. A paleomagnetic investigation of the Permo-Carboniferous Maroon and upper Permian – lower Triassic State Bridge Formations in north central Colorado. M.S. thesis, University of Texas at Dallas, Dallas.

Cole, F., Butler, R.F., and Gehrels, G.E. 1992. Paleomagnetism of late Paleozoic rocks in the northern Cache Creek terrane near Atlin, British Columbia. Canadian Journal of Earth Sciences, **29**: 486–498.

Davis, G.A., Monger, J.W.H., and Burchfiel, B.C. 1978. Mesozoic construction of the Cordilleran "collage", central British Columbia to central California. *In* Mesozoic Paleogeography of the Western United States. *Edited by* D.G. Howell and K.A. McDougall. Society of Economic Paleontologists and Mineralogists, Pacific Section, Los Angeles, pp. 1–32.

Dunlop, D.J. 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. Geophysical Journal of the Royal Astronomical Society, **27**: 37–55.

Ekstrand, E.J., and Butler, R.F. 1989. Paleomagnetism of the Moenave Formation: Implications for the Mesozoic North American apparent polar wander path. Geology, **17**: 245–248.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, A, **217**: 295–305.

Gabrielse, H. 1963. McDame map area, British Columbia. Geological Survey of Canada, Memoir 319.

Gabrielse, H. 1985. Major dextral transcurrent displacements along the Northern Rocky Mountain Trench and related lineaments in north-central British Columbia. Geological Society of America Bulletin, **96**: 1–14.

Gabrielse, H. 1991. Late Paleozoic and Mesozoic terrane interactions in north-central British Columbia. Canadian Journal of Earth Sciences, **28**: 947–957.

Globerman, B.R., and Irving, E. 1988. Mid-Cretaceous paleomagnetic reference field for North America: Restudy of 100 Ma intrusive rocks from Arkansas. Journal of Geophysical Research, B, **93**: 11 721 – 11 733.

Gordey, S.P., Gabrielse, H., and Orchard, M.J. 1982. Stratigraphy and structure of the Sylvester Allochthon, southwest McDame map area, northern British Columbia. *In* Current research, part B. Geological Survey of Canada, Paper 82-1B, pp. 101–106.

Gordon, R.G. 1984. A Paleocene North American paleomagnetic pole incorporating declination-only data. Geophysical Research Letters, **11**: 477–480.

Gunderson, J.A., and Sheriff, S.D. 1991. A new Late Cretaceous paleomagnetic pole from the Adel Mountains, west central Montana. Journal of Geophysical Research, B, **96**: 317–326.

Hagstrum, J.T., and Sedlock, R.L. 1992. Paleomagnetism of Mesozoic red chert from Cedros Island and the San Benito Islands, Baja California, Mexico revisited. Geophysical Research Letters, **19**: 329–332.

Hansen, V.L. 1990. Yukon–Tanana terrane: A partial acquittal. Geology, **18**: 365–369.

Harland, W.B., Armstrong, R.L., Cox, A.V., Craig, L.E., Smith, A.G., and Smith, D.G. 1990. A geologic time scale 1989. Cambridge University Press, New York.

Harms, T.A. 1985. Pre-emplacement thrust faulting in the Sylvester Allochthon, northeast Cry Lake map area, British Columbia. *In* Current research, part A. Geological Survey of Canada, Paper 85-1A, pp. 301–304.

Harms, T.A. 1986*a*. Structural and tectonic analysis of the Sylvester allochthon, northern British Columbia: Implications for paleogeography and accretion. Ph.D. thesis, University of Arizona, Tucson.

Harms, T.A. 1986*b*. The Sylvester Allochthon: A tectonic oceanic assemblage record of proto-Pacific telescoping and obduction. Geological Society of America, Abstracts with Programs, **18**: 114.

Harms, T.A., and Murchey, B.L. 1992. Setting and occurrence of Late Paleozoic radiolarians in the Sylvester allochthon, part of a proto-Pacific ocean floor terrane in the Canadian Cordillera. Palaeogeography, Palaeoclimatology, and Palaeoecology, **96**: 127–139.

Harmes, T.A., Nelson, J., and Bradford, J. 1988. Geological transect across the Sylvester allochthon north of the Blue River, Northern British Columbia (104P/12). British Columbia Ministry of Energy, Mines and Petroleum Resources, Paper 1988-1, pp. 245–248.

Helsley, C.E., and Steiner, M.B. 1974. Paleomagnetism of the Lower Triassic Moenkopi Formation. Geological Society of America Bulletin, **85**: 457–464.

Hesse, R. 1988. Origin of chert: Diagenesis of biogenic siliceous sediments. Geoscience Canada, **15**: 171–192.

Irving, E., and Monger, J.W.H. 1987. Preliminary paleomagnetic results from the Permian Asitka Group, British Columbia. Canadian Journal of Earth Sciences, **24**: 1490–1497.

Jones, D.L., and Murchey, B. 1986. Geologic significance of Paleozoic and Mesozoic radiolarian chert. Annual Reviews of Earth and Planetary Sciences, **14**: 455–492.

Kirschvink, J.L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Klepacki, D.W., and Wheeler, J.O. 1985. Stratigraphic and structural relations of the Milford, Kaslo and Slocan Groups, Goat Range, Lardeau and Nelson map areas, British Columbia. *In* Current research, part A. Geological Survey of Canada, Paper 85-1A, pp. 277–286.

Klepacki, D.W., Read, P.B., and Wheeler, J.O. 1985. Geology of the headwaters of Wilson Creek, Lardeau map area, southeastern

British Columbia. *In* Current research, part A. Geological Survey of Canada, Paper 85-1A, pp. 273–276.

Kluth, C.F., Butler, R.F., Harding, L.E., Shafiqullah, M., and Damon, P.E. 1982. Paleomagnetism of Late Jurassic rocks in the northern Canelo Hills, southeastern Arizona. Journal of Geophysical Research, B, **87**: 7079–7086.

Larochelle, A., and Currie, K.L. 1967. Paleomagnetic study of igneous rocks from the Manicouagan structure. Quebec. Journal of Geophysical Research, B, **72**: 4163–4169.

Lowrie, W., and Heller, F. 1982. Magnetic properties of marine limestones. Reviews of Geophysics and Space Physics, **20**: 171–192.

May, S.R., Butler, R.F., Shafiqullah, M., and Damon, P.E. 1986. Paleomagnetism of Jurassic rocks in the Patagonia Mountains, southeastern Arizona: Implications for the North American 170 Ma reference pole. Journal of Geophysical Research, B, **91**: 11 545 – 11 555.

McCabe, C., and Elmore, R.D. 1989. The occurrence and origin of Late Paleozoic remagnetization in the sedimentary rocks of North America. Reviews of Geophysics, **27**: 471–494.

McFadden, P.L., and Reid, A.B. 1982. Analysis of palaeomagnetic inclination data. Geophysical Journal of the Royal Astronomical Society, **69**: 307–319.

Miller, J.D., and Kent, D.V. 1988. Regional trends in the timing of Alleghenian remagnetization in the Appalachians. Geology, **16**: 588–591.

Monger, J.W.H. 1977. Upper Paleozoic rocks of the western Canadian Cordillera and their bearing on Cordilleran evolution. Canadian Journal of Earth Sciences, **14**: 1832–1859.

Monger, J.W.H. 1989. Overview of Cordilleran geology. *In* Western Canadian sedimentary basins. *Edited by* B.D. Ricketts. Canadian Society of Petroleum Geologists, Calgary, pp. 9–32.

Monger, J.W.H., and Berg, H.C. 1987. Lithotectonic terrane map of western Canada and southeastern Alaska. United States Geological Survey, Miscellaneous Field Studies Map MF-1874B.

Monger, J.W.H., and Ross, C.A. 1971. Distribution of fusulinaceans in the western Canadian Cordillera. Canadian Journal of Earth Sciences, **8**: 259–278.

Monger, J.W.H., Wheeler, J.O., Tipper, H.W., Gabrielse, H., Harms, T., Struik, L.C., Campbell, R.B., Dodds, C.J., Gehrels, G.E., and O'Brien, J. 1991. Upper Devonian to Middle Jurassic Assemblages. Part B. Cordilleran terranes. *In* Geology of the Cordilleran Orogen in Canada. *Edited by* H. Gabrielse and C.J. Yorath. Geological Survey of Canada, Geology of Canada, No. 4, pp. 281–327.

Murchey, B.L., and Jones, D.L. 1983. Tectonic and paleobiologic significance of Permian radiolarian distribution in Circum-Pacific region. American Association of Petroleum Geologists Bulletin, **67**: 522.

Murray, R.W., Jones, D.L., and Buchholtz ten Brink, M.R. 1992. Diagenetic formation of bedded chert: Evidence from chemistry of the chert–shale couplet. Geology, **20**: 271–274.

Nelson, J.L., and Bradford, J.A. 1987. Geology of the area around the Midway deposit northern British Columbia (104O/16). *In* Geological fieldwork 1986. British Columbia Ministry of Energy, Mines and Petroleum Resources, Paper 1987-1, pp. 181–192.

Nelson, J., and Bradford, J. 1989. Geology and mineral deposits of the Cassiar and McDame map areas, British Columbia (104P3/5). *In* Geological fieldwork 1988. British Columbia Ministry of Energy, Mines and Petroleum Resources, Paper 1989-1, pp. 323–338.

Nelson, J., Bradford, J., Green, K.C., and Marsden, H. 1988. Geology and patterns of mineralization, Blue Dome map area, Cassiar District (104P/12). *In* Geological fieldwork 1987. British Columbia Ministry of Energy, Mines and Petroleum Resources, Paper 1988-1, pp. 233–243.

Nelson, J., Ferri, F., and Schiarizza, P. 1989. Emplacement of the Slide Mountain terrane: obduction or shortening? Eos Transactions American Geophysical Union, **70**: 136.

Oliver, J. 1986. Fluids expelled tectonically from orogenic belts: their role in hydrocarbon migration and other geologic phenomena. Geology, **14**: 99–102.

Orchard, M.J., and Irwin, S. 1988. Conodont biostratigraphy, Midway property, northern British Columbia (104O/16). *In* Geological fieldwork 1987. British Columbia Ministry of Energy, Mines and Petroleum Resources, Paper 1988-1, pp. 249–253.

Plumley, P.W., Coe, R.S., and Byrne, T. 1983. Paleomagnetism of the Paleocene Ghost Rocks Formation, Prince William Terrane, Alaska. Tectonics, **2**: 295–314.

Pullaiah, G.E., Irving, E., Buchan, K.L., and Dunlop, D.J. 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters, **28**: 133–143.

Rejebian, V.A., Harris, A.G., and Huebner, J.S. 1987. Conodont color and textural alteration: An index to regional metamorphism, contact metamorphism, and hydrothermal alteration. Geological Society of America Bulletin, **99**: 471–479.

Richards, D.R., Butler, R.F., and Harms, T.A. 1991. Paleomagnetism of the Late Paleozoic Slide Mountain terrane, northern and central British Columbia. Eos Transactions, American Geophysical Union, **72**: 128.

Robertson, W.A. 1967. Manicouagan, Quebec, paleomagnetic results. Canadian Journal of Earth Sciences, **4**: 641–649.

Shibuya, H., and Sasajima, S. 1986. Paleomagnetism of red cherts: a case study in the Inuyama area, central Japan. Journal of Geophysical Research, B, **91**: 14 105 – 14 116.

Shive, P.N., Steiner, M.B., and Huycke, D.T. 1984. Magnetostratigraphy, paleomagnetism and remanence acquisition in the Triassic Chugwater Formation of Wyoming. Journal of Geophysical Research, B, **89**: 1801–1815.

Steiner, M.B., and Helsley, C.E. 1975. Reversal pattern and apparent polar wander for the Late Jurassic. Geological Society of America Bulletin, **86**: 1537–1543.

Struik, L.C. 1987. The ancient western North American margin: An alpine rift model for the east-central Canadian Cordillera. Geological Survey of Canada, Paper 87-15.

Struik, L.C. 1988. Structural geology of the Cariboo gold mining district, east-central British Columbia. Geological Survey of Canada, Memoir 421.

Struik, L.C., and Orchard, M.J. 1985. Late Paleozoic conodonts from ribbon chert delineate imbricate thrusts within the Antler Formation of the Slide Mountain terrane, central British Columbia. Geology, **13**: 794–798.

Toriumi, M. 1982. Strain, stress and uplift. Tectonics, **1**: 57–72.

Toriumi, M. 1986. The origin of strain patterns resulting from contemporaneous deformation and metamorphism in the Sambagawa metamorphic belt. Journal of Metamorphic Geology, **4**: 409–420.

Toriumi, M., Masui, M., and Mori, K. 1988. Strain in metacherts as determined from deformed shapes of radiolaria and Fry's method. Tectonophysics, **145**: 157–161.

Vandall, T.A., and Palmer, H.C. 1990. Canadian Cordilleran displacement: paleomagnetic results from the Early Jurassic Hazelton Group, Terrane I, British Columbia, Canada. Geophysical Journal of the Royal Astronomical Society, **103**: 609–619.

Van der Voo, R. 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions. Reviews of Geophysics, **28**: 167–206.

Wheeler, J.O., Brookfield, A.J., Gabrielse, H., Monger, J.W.H., Tipper, H.W., and Woodsworth, G.J. 1991. Terrane map of the Canadian Cordillera. Geological Survey of Canada, Map 1713A.